

solvers iterativos

$$a_{00}x_0 + a_{01}x_1 + a_{02}x_2 = b_0$$

$$a_{10}x_0 + a_{11}x_1 + a_{12}x_2 = b_1$$

$$a_{20}x_0 + a_{21}x_1 + a_{22}x_2 = b_2$$

$$x_0^{(k+1)} = \frac{b_0 - \frac{a_{01}x_1^{(k)}}{a_{00}} - \frac{a_{02}x_2^{(k)}}{a_{00}}}{1}$$

$$x_1^{(k+1)} = \frac{b_1 - \frac{a_{10}x_0^{(k)}}{a_{11}} - \frac{a_{12}x_2^{(k)}}{a_{11}}}{1}$$

$$x_2^{(k+1)} = \frac{b_2 - \frac{a_{20}x_0^{(k)}}{a_{22}} - \frac{a_{21}x_1^{(k)}}{a_{22}}}{1}$$

$$\begin{pmatrix} \textcircled{3} & -1 & 1 \\ 2 & \textcircled{5} & 2 \\ 1 & 0 & \textcircled{6} \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} \quad \begin{array}{l} |3| > |-1| + |1| \\ |5| > |2| + |2| \\ |6| > |1| + |0| \end{array}$$

$$|v| = \sqrt{v_1^2 + v_2^2 + \dots}$$

$$|v|^2 = v_1^2 + v_2^2 + \dots$$

metodo de Jacobi

$$AX = B \quad X = A^{-1}B$$

$$\begin{pmatrix} 2 & -1 \\ 4 & 6 \\ 2 & 3 \\ 5 & -1 \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 6 \\ 7 \end{pmatrix}$$

$$A \quad X = \quad B$$

$$X = A^* B$$

↳ pseudo inversa

$$\|A(X) - B\| = \|E\|_{\min}$$

Ecs normales

$$\begin{array}{ccc}
 A & \times & = B \\
 m \times n & n \times 1 & m \times 1 \\
 4 \times 2 & 2 \times 1 & 4 \times 1
 \end{array}$$

$$\underbrace{\begin{pmatrix} A^T & A \end{pmatrix}}_{\substack{n \times m \quad m \times n \quad n \times 1 \\ n \times n \quad n \times 1}} X = \underbrace{A^T B}_{\substack{n \times m \quad m \times 1 \\ n \times 1}}$$

$$X = (A^T A)^{-1} (A^T B)$$

$$X = \underbrace{\left[(A^T A)^{-1} A^T \right]}_{A^*} B$$